

Systems Thinking for AI Engineers

Dr Nawfal Al Hashimy

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Intended Learning Outcomes

By the end of this lecture, students will be able to:

1. Explain the principles of systems thinking and their relevance to AI engineering.
2. Distinguish systems thinking from system theory, systemic design, and systematic design.
3. Explain the nine foundational concepts of systems thinking (holism, reductionism, interconnectedness, boundaries, feedback loops, emergence, hierarchy, delays, adaptation) with examples.
4. Apply systems-thinking tools and mindsets to analyse a real-world case problem.
5. Identify leverage points for intervention in a complex system.

Agenda

- Vocabulary
- Why Systems Thinking?
- Foundations
- Case Example: Suzanne
- Conclusion

Vocabulary: Four Related Terms

① System Theory

Knowledge about how systems work (general knowledge).

① Systems Thinking

Diagnosing a *specific* problem using that knowledge.

① Systemic Design

Designing solutions that account for the whole system and its feedback loops.

① Systematic Design

Breaking design into orderly, methodical steps — doesn't by itself require systems thinking.

Introduction: Why Systems Thinking for AI?

AI engineers often focus on algorithms, data, or hardware in isolation. Yet AI systems rarely operate in isolation—they exist within dynamic, multi-stakeholder ecosystems. From autonomous vehicles to healthcare diagnostics, their success depends not only on technical accuracy but also on how they integrate with human, organisational, and societal contexts.

Systems thinking provides a framework for recognising interconnections, feedback loops, and emergent behaviour. Instead of viewing an AI model as a “black box,” systems thinking situates it within a web of dependencies: hardware, data flows, ethics, regulations, and end-users.

Three Tools of Systems Thinkers

① **Things *and* Connections**

Not just what a system is made of, but how its parts flow into each other — a university as a *network of knowledge flow*, not just departments and people.

① **Outflow, Not Just Inflow**

Asking not only how to acquire (talent, knowledge, resources) but how it's used, retained, or lost.

① **Cost of Inaction, Not Just Cost of Action**

Resources and risks accumulate invisibly (burnout, undocumented knowledge) until they cross a threshold — trends over time reveal this, a single snapshot doesn't.

Three Mindsets in Practice

① Zoom In and Out

Move between a specific pain point and the larger whole it sits in — a single household's food waste vs. the whole food ecosystem.

① Shift Perspective

View the problem through someone else's eyes, or borrow an analogy from an unrelated context — food waste reframed as "old clothing": discarded, but also donatable, repurposable.

① Be Aware of Your Own Lens

Question the assumptions your own background and expertise bring, before they quietly limit the problem you think you're solving.

Foundations of Systems Thinking

1. Holism
2. Reductionism
3. Interconnectedness
4. Boundaries
5. Feedback Loops
6. Emergence
7. Hierarchy and Nested Systems
8. Delays and Nonlinearity
9. Adaptation and Self-Organisation

Reductionism (recap)

Reductionism is an approach to understanding systems by breaking them down into their individual parts and analysing each part separately. The assumption is: if you understand all the parts, you understand the whole.

✓ Strengths

- Good for very focused, technical problems.
- Works well in controlled environments (e.g., debugging an algorithm).

✗ Limitations

- Misses how parts interact.
- Can't explain emergent behaviour (unexpected outcomes that arise from interactions).

☰ Example:

Fixing a car by looking at each part (engine, tires, brakes) separately.

Holism

Holism is an approach to understanding systems by looking at the whole system and its interactions, not just the individual parts. The assumption is: the whole is greater than the sum of its parts.

✓ Strengths

- Captures interconnectedness and feedback loops.
- Explains emergent properties (new behaviours that can't be predicted from individual parts).

✗ Limitations

- Can be harder to analyse because of complexity.
- Risk of being too broad or vague if not structured.

☰ Example :

Understanding that even if each part works perfectly, the car won't function unless the driver, traffic rules, and fuel supply are also considered.

Holism vs. Reductionism

Reductionism isolates components for analysis (e.g., testing one part of the fridge). Holism emphasises relationships and the bigger picture.

🔗 So

Reductionism = “focus on the pieces.”

Holism = “focus on the connections and the whole picture.”

📌 **In practice, engineers need both:**

- Reductionism to design and improve specific algorithms/components.
- Holism to ensure the system works responsibly within a larger context.

Interconnectedness

Every element in a system is connected; outcomes arise from these connections.

≡ Example:

Fixing a car by looking at each part (engine, tires, brakes) separately.

Boundaries

Defining what is “inside” and “outside” the system is essential, but boundaries are subjective and political.

≡ Example:

A school. Do you define the boundary as just classrooms and teachers? Or also include parents, neighbourhood, and online learning platforms? Where you “draw the line” changes how you see the system.

Feedback Loops

A feedback loop is when an action in a system produces an effect, and that effect goes back into the system and influences the original action. **Systems are regulated by reinforcing (positive) and balancing (negative) feedback loops.**

- They show dynamics over time (not just static cause–effect).
- They explain why systems don't always behave as expected.
- They help us spot leverage points — places where a small change breaks a harmful loop or strengthens a good one.

Feedback Loops as a Model of Mind

Michio Kaku's theory: consciousness is the sum total of all feedback loops needed to build a model of yourself in space, society, and time.

Unit of consciousness: a thermostat — one feedback loop (temperature) = one unit.

Evolution added feedback loops back-to-front in the brain — each layer modelling a new domain:

- **Hindbrain** (reptilian) → space
- **Limbic system** (mammalian) → society
- **Prefrontal cortex** (human) → time, via simulating the future

Level	Entity	Brain region	Domain modelled
0	Plants	—	Temperature, humidity, gravity, sprouting
1	Reptiles · Robots	Reptilian (hindbrain)	Space
2	Social mammals	Limbic system (midbrain)	Society
3	Humans	Prefrontal cortex (forebrain)	Time — simulating the future

Why Robots Stay at Level 1

Robots perceive their surroundings and react, but have no emotions and no sense of social hierarchy. They can simulate the future only in one narrow direction (e.g. airflow over a wing) — humans simulate the future *in all directions*.

Positive Loops

✓ Reinforcing (Positive) Loop

- The effect makes the original action stronger (snowball effect).
- It leads to growth, amplification, or escalation.

☰ Example – AI/Smart Fridge:

- Popular song → more people listen → charts go up → even more people discover it → popularity grows.

This is a reinforcing loop (R) because success breeds more success.

Balancing Loops

✗ Balancing (Negative) Loop

- The effect works against the original action (self-correcting).
- It leads to stability, limits, or resistance.

☰ Example – AI/Smart Fridge:

Thermostat → room gets cold → heating turns on → room warms → heating switches off → temperature stabilises.

This is a balancing loop (B) because the system resists too much change.

Emergence (Recap)

Systems display properties that cannot be predicted by analysing individual components.

Hierarchy and Nested Systems

Systems are organised in hierarchies; subsystems form part of larger systems.

Delays and Nonlinearity

Cause and effect may not be immediate, proportional, or predictable.

Adaptation and Self-Organisation

Many systems adapt and self-organise without central control.

Systemic design = applying that understanding to design practice.

Systemic Design Principles

Systems thinking tells us how systems behave. Systemic design is what happens when we apply that understanding to the practice of designing solutions:

Acknowledge Interconnections

Design decisions ripple across the whole system, not just the component you're touching.

Redefine Waste as a Resource

What looks like leftover output in one part of the system can be input somewhere else.

Design Self-Organising Systems

Build in the capacity for the system to adapt itself, rather than controlling every outcome centrally.

Use Feedback as a Guide

Treat feedback loops as the mechanism for adaptive improvement, not just as something to observe.

Case Example: Suzanne

Suzanne is a senior manager in a large Regional Health Organization. The high rate of obesity is an issue in her community, and she has been mandated to address it.

- **Her first instinct:** develop a program to get more people physically active.
- **The question:** what is realistic to expect from this approach alone?

Systems thinkers argue that treating a single program as an isolated *part*, without regard to the larger *whole* it operates within, tends to ignore other factors that influence its potential for impact.



Outcomes for Suzanne

Using a systems approach helps Suzanne:

- Focus on the broader issue of obesity in her community, not just a single program
- Better understand which external factors influence the program's success
- Set more reasonable expectations of what the program can accomplish
- Identify what needs to change both *within* and *outside* the program to maximize her organization's effectiveness

Conclusion

Systems thinking reframes problems — AI or otherwise — as the product of interacting parts, not isolated causes.

- Holism and reductionism are complementary, not opposed — engineers need both.
- Feedback loops, boundaries, and emergence explain why systems don't behave the way their parts alone would suggest.
- Zooming in/out, shifting perspective, and tracking leverage points turn these foundations into a practical toolkit — as the Suzanne case shows, a single program can't fix a problem that's systemic.

Many now believe a systems approach holds the most promise for addressing complex problems — in health, in AI, and beyond.